Diabetes Technology Expert Program

Automated Insulin Delivery Systems

1. Introduction

Welcome to this comprehensive module on automated insulin delivery or AID systems.

Automated insulin delivery systems are advanced insulin pumps that automatically release insulin every 5 to 10 minutes based on continuous glucose sensor readings and a sophisticated algorithm. While users still need to input carbohydrate information and exercise details, the pump dynamically adjusts insulin delivery to help control glucose levels. This module explores automated insulin delivery systems in detail, providing valuable insights into their benefits and usage.

Automated insulin delivery systems offer significant advantages over injections or conventional insulin pump therapy. They provide more time-in-range, with less hypoglycemia and lower glycemic variability. Additionally, automated insulin delivery systems ease the burden of diabetes management by improving sleep quality and reducing the number of daily care decisions involved in diabetes management.

With an increasing number of automated insulin delivery systems entering the market, it can be challenging for people with type 1 diabetes and their healthcare providers to keep up. Moreover, the relatively new nature of this technology raises questions about data interpretation and handling specific situations like hypo- and hyperglycemia, exercise, high-fat meals, swimming, illness, and travel.

The instructional videos featured in this module are presented by Dr. Inge Van Boxelaer, founder of Diabetotech and endocrinologist at the St-Lucas Hospital in Ghent in Belgium. The content has been reviewed by Dr. Liesbeth Van Huffel, endocrinologist at the OLV Hospital in Aalst in Belgium, and Melissa Holloway, a person living with type 1 diabetes and an advocate for people with diabetes from London, and Renate Groenendijk from the Netherlands, also a person living with type 1 diabetes supporting her peers.

All contributors to this module have no financial interests in medical technology firms, ensuring an impartial evaluation of the pros and cons of different automated insulin delivery systems.

Throughout the next hour, you will discover:
- How automated insulin delivery can be evaluated based on the CARES reference framework
- What education points are essential for users of AID systems
- Which automated insulin delivery is suitable for different user profiles
- How to interpret automated insulin delivery system reports in three simple steps
- And general guidelines for managing hypo- and hyperglycemia, dealing with high-fat meals, exercise, illness, alcohol and travelling.

By the end of this module, you will understand how to select the most appropriate automated insulin delivery system for yourself or your patients and effectively fine-tune the system to achieve optimal glycemic control. Best of luck on your journey to enhanced diabetes management!

2. The CARES framework

Welcome to this video where we will discuss how to compare different Automated Insulin Delivery or AID systems, using the CARES framework.

Automated insulin delivery systems have revolutionized insulin delivery for better diabetes management. To help healthcare providers compare different automated insulin delivery systems, the CARES reference framework provides valuable insights into their functionalities. The CARES framework, named for its focus on Calculate, Adjust, Revert, Educate, and Sensor/Share, encompasses the key considerations essential for evaluating these systems. By exploring the key aspects of the CARES framework, healthcare professionals can effectively assess and compare the functionalities of the most commonly used automated insulin delivery systems, enabling them to make informed decisions in optimizing diabetes care.

Let's explore the key aspects of the CARES framework for the most commonly used automated insulin delivery systems:

2.1 Calculate

- In the MiniMed 780G, the insulin delivery is calculated every 5 minutes based on the total daily insulin dose of the last 2-6 days.
- In Control-IQ technology used in Tandem pumps, the preset basal insulin rate is modulated based on the predicted glycemia, total daily insulin dose and weight.
- The mylife Loop AID system delivers extended insulin boluses, calculated using predicted glycemia, total daily insulin dose, and weight.
- The Diabeloop algorithm calculates microboluses based on predicted glycemia, total daily insulin dose, weight, and average carbohydrate intake.
- Omnipod 5 delivers microboluses based on predicted glycemia and the total insulin dose of the past three days.
- The iLet from Beta Bionics calculates its automated insulin delivery solely based on weight.
- The TouchCare Nano system delivers insulin every 2 minutes based on previous, current and predicted glucose and insulin delivery.
- Finally, in the twiist AID system and open-source AID systems like AndroidAPS, DIY Loop, iAPS and Trio, the preset basal insulin rate is modulated based on predicted glycemia, and insulin and carbohydrates on board.

2.2 Adjust

- Each system allows adjustment of different parameters such as target value, basal insulin rate, carbohydrate ratio, and insulin sensitivity. However, specific limitations may apply to certain parameters for each system.

2.3 Revert

- Automated Insulin delivery systems generally revert to manual mode if there is no sensor data or connection with the insulin pump for 20 to 30 minutes.
- Reverting to manual mode may also occur due to sensor warm-up or concerns about sensor integrity.

2.4 Educate

General education points for automated insulin delivery systems include managing hypoglycemia, recognizing infusion set issues, trusting the system's decision-making, proper timing of meal boluses, and responding to alarms. In the

next lesson, we will delve into these general education points in more detail, providing you with comprehensive knowledge to effectively utilize automated insulin delivery systems.
It's also important to note that each automated insulin delivery system may have specific educational considerations. These can include adjusting basal profiles for different activities or utilizing additional functions designed for specific situations. System-specific education points will be discussed in the respective system-specific modules.

2.5 Sensor

To ensure the proper functioning of an automated insulin delivery system, accurate sensors and continuous readings are vital.

- While the used sensors are reliable, occasional inaccuracies can occur, especially during the first 24 hours. If the sensor glucose values do not align with your expectations or if other signs indicate inaccuracy, it is important to perform a finger prick test. Make sure your hands are clean. If there is more than 20% difference between the sensor and finger prick readings, despite stable blood glucose within the target range, it is recommended to stop auto mode and calibrate the sensor if possible, or replace the sensor.

- If recurring sensor errors occur, try selecting a different body site to insert your sensor.
- Bluetooth connection may be disturbed by other nearby Bluetooth devices such as blood glucose meters, headsets, tablets or kitchen devices such as microwave ovens or ceramic hobs. In this case your receiver does not display any blood glucose values. When the bluetooth connection is reestablished, the data is backfilled.

2.6 Share

Most AID systems can share the system data, allowing healthcare providers and family members to monitor insulin and glucose data in real-time. However, the extent of data sharing and remote control capabilities differ across systems.

The evolution in automated insulin delivery systems goes really fast. This content is updated in November 2025. For the latest updates, please visit the company's websites. Additionally, staying updated with resources like the Diabetotech blog can help you explore the evolving diabetes tech field.

In conclusion, understanding the CARES reference framework empowers healthcare providers and students alike to navigate the ever-evolving landscape of automated insulin delivery systems with confidence. By acquiring knowledge about the Calculate, Adjust, Revert, Educate, and Sensor/Share aspects, you can develop a comprehensive understanding of these systems and their potential benefits for diabetes management. This knowledge equips you to adapt to new technologies, make informed decisions, and provide optimal care to individuals with diabetes.

3. Key education points

Welcome to this video where we will discuss 10 key education points for users of Automated Insulin Delivery systems.

To ensure a smooth and effective transition into an automated insulin delivery system, it is important to understand both general education points and system-specific guidelines. Here are the essential aspects to cover in the general education.

1.Eat less carbs if hypoglycemia

In AID systems, mild hypoglycemia usually requires only 5 to 10 grams of carbohydrates for treatment, as opposed to traditional insulin therapy. This is because automated insulin delivery systems adjust the basal insulin rate before you even experience a hypo, resulting in less insulin on board, and consequently, a reduced need for carbohydrates to treat the hypoglycemic episode.

2. If in doubt, change it out!

If blood glucose levels remain high for an extended period without an apparent reason (for example more than 270 mg/dl or 15 mmol/l with no decreasing trend or no 50 mg/dl or 2.7 mmol/l decrease within one hour after an insulin bolus), consider the possibility of a blockage or infusion set problem. When in doubt, it is recommended to replace the tubing and infusion set, or the patch pump, promptly. The mantra is: “if in doubt, change it out”

In case of sustained hyperglycemia, it is also recommended to measure blood glucose with a finger prick test and monitor ketones.

A user-initiated correction bolus might be needed.

A correction may be more effective using an insulin pen, as in case of set failure and multiple correction attempts, the insulin on board may be falsely elevated thus preventing an additional bolus.

3. Trust the automated insulin delivery system

It is crucial to allow the AID system time to adjust and regulate blood sugar levels. Avoid reacting to every rise in glucose levels. Trust the system and let the algorithm increase automatic basal insulin delivery gradually, thereby slowly reducing blood glucose levels.

If a correction bolus is necessary, rely on the bolus calculator and avoid overriding the system unless there is suspicion of a blockage.

4. Bolus before meals

If using a rapid-acting insulin analogue like NovoRapid (Novolog) or Humalog, administer the meal bolus approximately 15 minutes before meals, unless managing hypoglycemia, gastroparesis, or consuming a high-fat meal, in which case it is appropriate to give the bolus directly before the meal.

If using ultra-fast-acting insulin like Fiasp or Lyumjev, administering the bolus 5-10 minutes before eating is sufficient.

When unsure of the exact carbohydrate intake, enter only the amount of carbohydrates you are certain of consuming.

It is important to give mealtime insulin before the carbohydrate intake. Bolusing after meals can lead to insulin stacking and potential hypoglycemia caused by the automatic increase in algorithm-modulated insulin delivery after an initial rise in glucose levels because of the meal itself.

In situations where a meal bolus is missed or delayed, consider giving half the bolus within 30-60 minutes after the start of the meal. If more than 60 minutes have passed, administer only a user-initiated correction bolus based on the glycemic rise.

Additionally, adults using automated insulin delivery systems may benefit from a low-carbohydrate diet to further improve glycemic control. For low-carb meals, consider counting proteins and fats as well. We recommend not eating less than 60 grams of carbs per day, as real ketogenic diets put you at risk for ketoacidosis.

5. Start Exercise Mode 90 minutes before activity

Set a higher glucose target by using Exercise Mode, ideally well before starting the activity (up to 1 to 2 hours in advance), particularly for prolonged aerobic exercise. This temporary target can be cancelled at the end of exercise or be maintained post exercise if post-exercise hypoglycemia is a concern.

The goal is to minimize activity at times of peak bolus insulin action. To facilitate this, reduce the pre-meal bolus dose by approximately 25 to 33% if prolonged exercise is anticipated within 2 hours of a meal. The extent of reduction in the meal bolus depends on various factors such as the type, intensity, and duration of the activity, your fitness level, current blood glucose level, and insulin on board.

6. Suspend insulin when disconnected

If the pump is disconnected for more than 15 minutes, it is important to stop insulin delivery to prevent miscalculations by the algorithm.

7. Respond to alarms

Respond to system alarms promptly by checking the device to identify the issue and follow the recommended advice.

Setting alarms of your automated insulin delivery system to silent or vibrate-only mode is strongly discouraged. These alarms are designed to notify you of important events and situations related to your diabetes management. By keeping your alarms audible and attention-grabbing, you can ensure that you promptly respond to any alerts and take appropriate actions to maintain your safety and well-being.

To avoid alarm fatigue, it is advised to minimize alarms to those that require immediate attention. One suggestion is to start with hypoglycemia alerts only (for example at 70 mg/dl or 3.9 mmol/l) and add hyperglycemia alerts if tolerated.

8. Disable automatic updates

Disable automatic updates on mobile devices if you're using any automated insulin delivery system apps. Automated insulin delivery system apps may experience compatibility issues after updates from Android or iOS. It is advisable to perform updates only when notified by the device manufacturer that the update is compatible.

9. Seek peer support

Seek peer support by joining Facebook groups and Discord chats and connecting with others using automated insulin delivery systems. Sharing experiences and learning from each other can provide valuable insights and a quick helpline for addressing issues. Remember, you are not alone in encountering practical challenges.

10. Set realistic expectations

To ensure a smooth transition to automated insulin delivery systems, it's essential to set realistic expectations and assess the user's willingness to adapt to new technologies.

It's important to note that automated insulin delivery systems require user interaction to achieve optimal results. Here are two essential actions to take:

- Enter Carbohydrate Intake: Before each meal, it's crucial to input the number of grams of carbohydrates you plan to eat into the system. This helps the system deliver a bolus to cover your meal effectively.
- Activate Exercise Mode: Make sure to use the Exercise Mode feature on your automated insulin delivery system 1 to 2 hours before you engage in physical activity. This allows the system to adjust insulin delivery accordingly, considering the increased energy expenditure during exercise.

The user interaction with AID systems plays a vital role in optimizing the performance of the system and achieving better glycemic control.

Embracing the journey of utilizing an automated insulin delivery system requires a solid foundation of knowledge and understanding. Remember, you are not alone in this journey, and with the right education and support, you can thrive with the advancements offered by automated insulin delivery systems.

4. Starting an AID System: Indications, Timing, and Choosing the Right System

Welcome to this video where will explore different topics around starting an automated insulin delivery system.

We will discuss the indications for AID systems, the optimal timing for initiating them, and the process of selecting the most suitable system for your needs.

4.1 Indications

Automated insulin delivery systems are recommended for all individuals with type 1 diabetes and insulin-deficient diabetes. According to the latest guidelines of the American Diabetes Association, AID systems should also be recommended in people with type 2 diabetes on multiple daily injections.

While the initial focus was on young, easily educable individuals who could adhere to carbohydrate counting, it has been shown that automated insulin delivery systems can also be beneficial for those with high HbA1c, older individuals, and those who struggle with consistent bolusing or carbohydrate counting.

The use of automated insulin delivery systems in very young children is still being studied, with promising preliminary results. Here you can see the age approvals for the different automated insulin delivery systems.

Most automated insulin delivery systems also have minimum requirements for total daily insulin dose and sometimes also for weight.

It is important for users to have sufficient vision and hearing or have a trained caregiver who can operate the system.

Currently, CamAPS FX and MiniMed 780G are the only automated insulin delivery systems approved for use in pregnancy, in Europe. Other systems are used off-label.

Automated insulin delivery system use during dialysis is not specified in the indications but is a topic of discussion.

Some automated insulin delivery systems require a compatible mobile phone, such as mylife Loop, Tandem Control-IQ with the Mobi Pump and the twiist AID system. Open-source automated insulin delivery systems require a computer for app development and periodic updates.

4.2 Timing of initiation

Early initiation of diabetes technologies in recently diagnosed individuals has shown improved and sustained long-term glycemic control, potentially reducing diabetes-related complications. Early initiation leads to better device acceptance and has particular advantages for preschool-children.

The potential of early initiation of automated insulin delivery systems to preserve beta cell mass is a promising area of research, although conclusive evidence is still needed to confirm its effectiveness.

4.3 Choosing the right system

While it is generally believed that all automated insulin delivery systems are equally effective in glycemic control, some researchers suggest that the MiniMed 780G algorithm may be the most responsive algorithm, with slightly higher time in range (TIR) observed in real-world studies. However, the lack of large head-to-head studies comparing different systems and involving different target groups makes it challenging to substantiate these claims.

The main differences among automated insulin delivery systems lie in

- the appearance and size of the insulin pump and CGM,
- ease of use,
- phone control options,
- target value settings,
- and the degree and method of interaction with the algorithm.

Choosing an automated insulin delivery system is typically done in consultation with a healthcare provider, considering factors such as reimbursement, availability at the diabetes center or nearby, and specific advantages and disadvantages of each system.

Comparison websites can assist individuals in making informed choices based on their preferences and lifestyle. People with diabetes should have the opportunity to assess the benefits and burdens of available automated insulin delivery systems to determine the most suitable device for their needs.
Educational support, personal resources, age, availability, insurance, and personal preferences should all be considered, and unbiased sources should be heavily relied upon. The goal is to empower individuals with diabetes to make well-informed decisions about their automated insulin delivery system, ensuring the best possible fit for their lifestyle and management needs.

5. Interpreting reports

Welcome to this video on creating and interpreting reports for automated insulin delivery systems.

We will explore types of reports available, and provide a structured approach to interpreting them.

Please note that the information shared represents expert opinions rather than official guidelines.

5.1 Types of reports

Currently, there are no standardized reports for automated insulin delivery systems, resulting in variations across platforms such as CareLink, Glooko, Tandem Source, Yourloops, Nightscout and Tidepool.

To gain valuable insights, it is recommended to review reports from the past 14 days, similar to CGM reports.

Generally, there are three main types of reports to consider:

- * 1. Summary Report: This report provides an overview of glucose and insulin metrics. Important glucose sensor metrics include time of sensor wear, Time in Range (TIR), Time Below Range (TBR), estimated HbA1c, and glycemic variation. In the case of pregnancy, it is important to adjust the target range accordingly. The report usually contains an AGP profile that displays trends of hypo- or hyperglycemia, enabling you to identify patterns and assess the impact of meal boluses. The summary report also highlights details about the insulin delivery, like time in automode, total daily insulin dose, percentage of basal and bolus insulin and the frequency of bolusing.
- * 2. Daily & Weekly Reports: These reports offer a detailed view of the relationship between glucose levels and meal/correction boluses for each day.
- * 3. Device Settings Report: These reports show the settings and alarms of the glucose sensor, insulin pump and AID algorithm.

5.2 Four steps to interpret reports

To effectively interpret reports from automated insulin delivery systems, it is beneficial to follow a structured approach. Consider the following steps as a guide:

- 1. Assess Glycemic Information: Begin by evaluating if treatment goals are being met, focusing on Time In Range and above (>70%) and Time Below Range below (<4%) over the past two weeks.

    - o One important aspect to take into account when working with automated insulin delivery systems is to assess the frequency of meal bolusing by the user, which is typically indicated in the summary report.

- 2. Optimize Automated Insulin Delivery Settings: Analyze the AGP profile to identify trends of hypo- or hyperglycemia and check if they are related to meal boluses on the daily or weekly reports.

    - o If these trends are related to meal boluses, assess factors such as timing, carbohydrate counting, and consider adjusting the carbohydrate ratio. The goal is to minimize post-meal glucose excursions to < 60 mg/dL (**or** <3.3 mmol/l) compared to pre-meal.

  - For trends outside of meals, adjustments to the target or basal insulin rates may be necessary based on the automated insulin delivery system. Additionally, explore possible correlations with missed boluses, exercise, alcohol or hypoglycemia.

- 3. Guide Behavioural Recommendations: Ensure proper usage of the automated insulin delivery system by checking sensor wear, activation of auto mode, adherence to pre-meal bolusing, correct use of temporary target values, and regular infusion set changes. Evaluate daily reports for overcorrection of hypo- or hyperglycemia, monitor the frequency of manual boluses and alarms, and assess the appropriateness of alarm settings and reminders.

- 4. Review Pump Settings and Emergency Plans:

  - Verify that the preset basal insulin rate corresponds to the actual basal insulin in auto mode. If there is a discrepancy, it is recommended to make adjustments to the basal rate by 10% in each time block, rather than making individual adjustments per time block.
  - Document all pump settings and establish an emergency plan, including insulin pen usage, in the event of a pump malfunction. Do not forget to ensure that your patient has access to insulin pens.

5.3 Do's & don'ts

- Implement changes to insulin settings gradually (1 or 2 at a time) and evaluate their effect on glucose control after 1-2 weeks.
- Printing the summary report and noting down the goals and changes discussed during the consultation can be beneficial. This practice helps empower patients to better understand their data.
- As a user of an automated insulin delivery system, it is best to take an active role in reviewing personal reports and considering adjustments when necessary, in addition to consultations with healthcare providers.

Understanding and effectively interpreting reports for automated insulin delivery systems are vital for optimizing glucose control and overall diabetes management. By following a structured approach and collaborating with healthcare providers, users can make informed adjustments, monitor trends, and enhance their journey toward better glucose management and well-being.

Remember, managing your diabetes is a continuous process, and with diligent monitoring and proactive engagement with your reports, you are taking significant steps toward achieving better glucose control and overall well-being.

6. Managing Special Situations with AID Systems: Hypo/Hyperglycemia, High-fat Meals, Exercise, Illness, Alcohol and Travel

In this video, we will delve into the specific considerations and practical advice for effectively managing various situations when using automated insulin delivery systems.

These situations include dealing with hypoglycemia, hyperglycemia, high-fat meals, exercise, illness, alcohol and travel.

- It is important to note that the following guidelines are expert opinions and not official recommendations.
- Be aware that everybody is unique and your diabetes may vary.
- On top of that, each automated insulin delivery system has its unique features, so consulting healthcare providers and adhering to personalized plans is vital for effective management.

6.1 Hypoglycemia

With automated insulin delivery systems, fewer carbohydrates are typically needed to treat hypoglycemia compared to traditional insulin therapy.

- During a hypoglycemic event, it is recommended to consume only 5-10 grams of “uncovered” carbohydrates, unless exercising or when a significant meal bolus overestimation occurs.

- Certain automated insulin delivery systems, like CamAPS FX, Diabeloop, twiist and open-source automated insulin delivery systems, allow you to enter the carbohydrates you consume to treat hypoglycemia without administering insulin for them. It is recommended to follow this approach so that the system's algorithm can account for these carbohydrates.

- Wait for 15 minutes before re-treating hypoglycemia to avoid oscillating glucose levels. After a correction for hypoglycemia, consider performing a finger prick test to assess the need for further sugar intake, since sensor readings may lag.

To prevent hypoglycemia, you can temporarily increase the target value on your automated insulin delivery system. You might find this useful in the following situations: during a period of recurring hypoglycemia, or during an active holiday or in a warmer climate, when starting a new job or returning to school or training after holidays

If someone using an automated insulin delivery system experiences a severe hypoglycemic episode with reduced consciousness or enters a coma, the response should be the same as with someone without an automated insulin delivery system.

- Encourage them to consume sugar, and if that fails, administer glucagon and call an ambulance.
- In the case of someone in a hypo coma, you may disconnect the insulin pump since they do not need additional insulin. Be sure not to misplace the device.
- After administering glucagon, the person should recover quickly.

6.2 Hyperglycemia

Hyperglycemia can occur when a person with diabetes is ill, stressed, in pain, taking corticosteroids, or experiencing hormonal changes (e.g., premenstrual period in women or a growth spurt in children).

In cases of mild hyperglycemia, it is generally recommended to rely on the automated insulin delivery system and avoid giving excessive manual correction boluses or false carbs, as this can often lead to hypoglycemia 1-3 hours later. If a manual correction bolus is necessary, it is advisable to use the bolus calculator whenever possible.

Some automated insulin delivery systems offer additional options for delivering more insulin, such as the Boost function in the CamAPS FX algorithm, increased aggressiveness for normo- and hyperglycemia in Diabeloop, the Pre-Meal Preset in twiist Loop or temporary target adjustments or increased insulin release percentage in open-source automated insulin delivery systems.

If blood glucose levels remain high for an extended period without an apparent reason (e.g., >270 mg/dl or 15 mmol/l with no decreasing trend or no 50 mg/dl or 2.7 mmol/l decrease within one hour after an insulin bolus), consider the possibility of a blockage or infusion set problem. When in doubt, it is recommended to replace the catheter and infusion set promptly. The mantra is: “if in doubt, change it out”

Anyone using an insulin pump should be familiar with what to do in case of hyperglycemia. Various step-by-step plans are available, depending on whether you have access to ketone strips or not. Here is an example:

- Monitor blood glucose with a finger prick test and ketones every 2-4 hours until the glycemia is back to normal. Elevated ketone levels indicate the need to switch to manual mode and follow the guidelines provided by your diabetes care team. In adults, ketones are considered elevated when they are higher than 1.5 mmol/L, while for children, this applies from 0.6 mmol/L. Once ketone levels return to normal, auto mode can be re-enabled.
- Administer a manual insulin correction bolus. If blood glucose levels do not decrease after a manual correction bolus, it is recommended to administer an insulin bolus using an insulin pen.
- Check and replace the infusion set, tubing, and insulin reservoir.
- Drink an adequate amount of water (e.g., 500 ml per hour). Hyperglycemia can lead to dehydration, which exacerbates the condition. It is important to drink plenty of fluids during hyperglycemic episodes.
- And be aware of the signs of ketoacidosis, such as vomiting. Vomiting is a sign of ketoacidosis. If persistent hyperglycemia causes vomiting, it is crucial to contact your diabetes care team or go to the emergency department for intravenous treatment with fluids and insulin.

Since a pump blockage can occur at any time, it is important to carry the necessary equipment to address such situations. This includes an insulin pump cartridge, infusion set, blood glucose meter, test strips, and an insulin pen with short-acting insulin to administer boluses if needed. When travelling, it is also recommended to have long-acting insulin available in case the pump malfunctions.

To avoid infusion set issues, it is advisable to change the infusion set every 2 to 7 days as recommended for your specific infusion set, and set a reminder on your pump for this task.

6.3 High-fat meals

High-fat meals pose challenges in insulin dosing due to delayed carbohydrate absorption.

Here are some general guidelines, although they may not work as well in everybody and need to be individualized:

- Enter the grams of carbs you are going to eat without taking into account the fats and proteins. If your automated insulin delivery system allows it, you can opt for an extended bolus.

- If you experience hypoglycemia shortly after the bolus, you can try administering only 70% of the carbohydrates and allow the automated insulin delivery system to compensate for the remaining glucose increase.

- If the previous approach doesn't work, you can try splitting the bolus by giving 50% of the carbohydrate grams before the meal and the remaining 50% two hours later.

Some automated insulin delivery systems provide additional features that can be used for a high-fat meal:

- Tandem pumps with Control-IQ technology allow you to change a normal meal bolus into an extended bolus. Twiist Loop and DIY Loop allows you to set the carb absorption time of the carbs you enter.
- CamAPS FX, Diabeloop, and AndroidAPS have an “Add carbs” option. This allows you to enter carbs that will be covered with insulin only when blood glucose levels rise.

6.4 Exercise

Physical activity increases the risk of hypoglycemia. Here are some tips to consider before, during, and after exercise:

- Before exercise:

    - Use the Exercise Mode of your automated insulin delivery system to reduce the risk of hypoglycemia during planned activities. Set it up 1 to 2 hours before and during the exercise to decrease insulin delivery while exercising.

- If you exercise within 2 hours of a meal, consider reducing the meal bolus by 25-33%, depending on the type, intensity, and duration of the activity, as well as your fitness level, current blood glucose level, and insulin on board.
- Unless trending towards hypoglycemia, avoid consuming carbohydrates 15 minutes to 1 hour before exercise. The automated insulin delivery system will automatically increase insulin delivery in response to carbs.
- Aim for a glucose >120 mg/dl or 6.7 mmol/l before starting moderate intensity aerobic exercise. Check your glucose 10 minutes before the start. If it's below 120 mg/dL or 6.7 mmol/L, consider consuming 15 grams of carbs.

- During exercise:

    - Use the Exercise Mode of your automated insulin delivery system to elevate the target for blood glucose levels.
    - Aim for a blood glucose level above 120 mg/dL or 6.7 mmol/L at the start and during the activity. If needed, consume small amounts of carbs gradually, for example, every 30 minutes, without entering them into the automated insulin delivery system.
    - You can try connecting your sensor app to your smartwatch or cycling computer. That way, you can more easily eat small amounts of carbohydrates from the moment your trend arrow tilts or your glycemia drops below 120 mg/dl (or 6.7 mmol/l).
    - If you need to disconnect the pump for more than 15 minutes (e.g., swimming), it's important to stop insulin delivery.
    - For prolonged exercise, it is advisable to maintain a low basal insulin rate and supplement with carbohydrates to prevent ketone development.

- If you experience hypoglycemia after exercising:

    - Continue using the Exercise Mode for 6 hours after exercise or throughout the night.

  - Consider reducing boluses by up to 50% for the post-exercise meal or consume a bedtime snack without entering it into the automated insulin delivery system.

Of course, these recommendations are general guidelines. Adjusting insulin dosages based on exercise trends and individual needs often requires a trial-and-error approach. It's essential to work closely with your healthcare team to fine-tune your insulin regimen for optimal management during physical activity.

6.5 Illness and medical procedures

During illness, it is common to experience hyperglycemia. In such cases, follow the guidelines for managing hyperglycemia.

When undergoing medical procedures or tests, it is generally best to disconnect the pump. However, the specific recommendations may vary depending on the type of procedure:

- For an X-ray or CT scan, you can leave the sensor and transmitter in place, ensuring they are away from the radiation field and if possible, covered with a lead apron.
- For an MRI, remove the sensor and transmitter before the procedure. If you use a patch pump or infusion set with a steel cannula, it is crucial to remove these as well, as contact between metal and skin during the MRI can cause burns. An exception is the FreeStyle Libre 2 and 3 sensors that are FDA-approved for use during MRI.

Upon admission to a hospital, if you are mentally fit, you may be able to continue using your automated insulin delivery system. However, this depends on the hospital's policy and should be discussed with your healthcare provider.

For endoscopy or minor procedures without diathermy, it may be possible to keep the automated insulin delivery system in place if agreed upon by the surgical team. In such cases, consider using a higher temporary target and ensure the infusion set and sensor placement are away from the surgical field. Keep your mobile phone or handset nearby if the algorithm runs on it.

For larger procedures with a duration of more than 2 hours, involving diathermy or when the surgical team does not agree, remove the pump and sensor, and switch off auto mode.

Please exercise caution with the accuracy of the sensor in cases of significant edema or surgeries with important hemodynamic impact. Additionally, be mindful of frequent intravenous use of paracetamol or acetaminophen during hospital admissions. These factors may affect the reliability of the sensor readings, and extra attention should be given to ensure accurate insulin management during such situations.

If the sensor remains in place during an X-ray, CT scan, or surgery with diathermy, it is recommended to check its accuracy afterward by performing a finger prick test. Calibrating the sensor may be necessary, and is possible for Dexcom and Medtronic sensors.

6.6 Alcohol

Alcohol can affect blood glucose levels, and caution should be exercised to prevent hypoglycemia.

Different types of alcoholic beverages can vary in their effects:

- Some alcoholic drinks, such as certain beers and breezers, contain carbohydrates that can cause a rise in blood glucose levels.
- Other drinks, like wine or spirits mixed with diet soda, may not contain carbohydrates.
- It's important to note that hypos can still occur hours later, irrespective of the initial effect on blood glucose levels.

Automated insulin delivery systems respond to rising glucose levels by increasing insulin delivery. However, it is unaware of alcohol consumption and may not anticipate the delayed lowering effect.

If you choose to drink alcohol, consider increasing your target level to reduce insulin administration. Consuming more carbohydrates will not help, as the automated insulin delivery system will compensate by increasing insulin delivery.

6.7 Travel Considerations

When travelling with an automated insulin delivery system, it's important to be prepared and take certain precautions. Here are some tips to keep in mind:

- Pack Sufficient Supplies: Make sure to pack an ample supply of all the necessary equipment and supplies for your automated insulin delivery system, including insulin, infusion sets, reservoirs, and glucose monitoring supplies. It's better to have more than you think you'll need to avoid any shortage during your trip. Don't forget your device charger and an appropriate travel plug if needed.

- It's always wise to carry an emergency card stating that you have diabetes.
- Have the right certificates: If you are flying, you might need a flight certificate for your insulin, sensor, and pump. Consider declaring medical baggage, which allows for an additional 2 kilograms of weight and ensures that your insulin and equipment won't be placed in the cargo hold where freezing may occur.

- Plan for Pump Failure: In case your pump malfunctions or fails during your trip, many pump manufacturers provide an emergency replacement service, delivering a new insulin pump to you within 24 hours. However, be aware that in remote or isolated locations, like islands, the delivery time may be longer. Therefore, it is essential to have backup plans in place. Some manufacturers also offer a holiday loaner pump, which can be ordered a few weeks before your trip, ensuring you have a spare pump for added peace of mind during your travels.

- Adjust Settings for Travel Activities: Consider using a higher target value for the first few days of your holiday. This can help prevent the risk of hypoglycemia, especially if you anticipate being more active than usual or travelling to a destination with warmer weather.

- Time Zone Adjustment: Most automated insulin delivery systems powered by a mobile phone app will automatically adjust the pump clock time when travelling across time zones. However, for specific pumps like MiniMed 780G and Tandem pumps, manual adjustment is necessary.

- Airplane Mode and Bluetooth Communication: Some airlines require passengers to put their mobile phones on airplane mode during take-off and landing, which can temporarily interrupt Bluetooth communication with your automated insulin delivery system. If this happens, don't worry. You can have a device in airplane mode and switch the Bluetooth back on separately. If this is not possible, after 30 minutes, the system will switch to the manual mode until Bluetooth connectivity is restored. Once you're allowed to use your phone again, remember to turn off airplane mode promptly to ensure auto mode resumes.

By following these tips, you can have a more seamless and stress-free experience while travelling with your automated insulin delivery system. Enjoy your trip and stay prepared for optimal diabetes management.

As you manage special situations with automated insulin delivery systems, keep in mind that achieving time in range is essential, but equally important is your time in happiness! Please take care of your emotional well-being and try to avoid diabetes burnout. Remember, diabetes management can be challenging, and it's okay to seek help and take breaks when needed. By staying informed, prepared, and mindful of your mental health, you can confidently navigate the complexities of diabetes management with automated insulin delivery systems and lead a fulfilling life.

7. A comprehensive automated insulin delivery overview

Congratulations on completing this module! You have gained valuable insights into the world of automated insulin delivery systems for diabetes management. In this final chapter, we will provide you with a thorough overview of the various automated insulin delivery systems available, highlighting their features, advantages, and considerations.

Having a comprehensive understanding of the different automated insulin delivery systems empowers you to make informed decisions about your diabetes management. Each system has its unique characteristics, functionalities, and compatibility requirements. By exploring this overview, you can identify the system that best aligns with your needs, preferences, and lifestyle.

Let's dive into the details of the different automated insulin delivery systems:

1. MiniMed 780G:

- Features: the MiniMed 780G has a very good algorithm with autocorrection boluses every 5 minutes and with a high time in range in pivotal and real-world studies. It is the only system that offers a 7-day infusion set.
- Considerations: Drawbacks are that the Guardian sensor might not last the full 7 days, and takes some time to replace, although Medtronic also offers a Simplera Sync and Instinct sensor which are much easier to use. The MiniMed pump itself is also quite large.

2. Tandem with Control-IQ technology:

- Features: Tandem pumps with Control-IQ technology allow for more user interaction as different profiles can be set and more parameters can be adjusted. It is one of the most used AID systems in the US and is a more modern-looking pump. Additionally, Tandem offers the Tandem Mobi which is only half the size of the t:slim X2 insulin pump and can be worn on the body.
- Considerations: Drawbacks are that there is no dedicated follow app available.

3. CamAPS FX:

- Features: The CamAPS algorithm is fully phone controlled. It is one of the only AID algorithms that is approved in pregnancy and it allows to set the glucose target value as low as 80 mg/dl or 4.4 mmol/l.
- Considerations: It's important to note that CamAPS FX requires you to have a compatible mobile phone and keep that phone close to your body at all times.

4. Diabeloop:

- Features: The DBLG1 algorithm is known for its many adjustable settings, its optional Zen-mode and its self-learning algorithm.
- Considerations: Drawbacks are that you need to have the handset close to your body at all times to stay in automode, and that there is no dedicated follow-app available.

5. Omnipod 5:

- Features: Omnipod 5 stands out as the first fully phone-controlled AID system with a pure patchpump and features an AID algorithm located on the pod

- Considerations: It's important to note that the Omnipod 5 mobile app is only available in the United States. Users outside the US must carry the Omnipod 5 Controller to operate their system. A dedicated follow app is not available.

6. iLet from beta Bionics
- Features: The iLet from Beta Bionics is known for its ease of use for the user and for the healthcare provider, as it can start in automode just by entering your weight. It works with meal announcements, so carb counting is not necessary.
- Considerations: Drawbacks are that the system is only available in the United States , the slightly less time in range results and the rather bulky insulin pump.

7. TouchCare Nano System
- Features: The TouchCare Nano system is an AID system with the smallest on-body devices. The algorithm runs on the pump, allowing the system to stay in automode even when the phone is not nearby. Its AI-based auto-meal feature enables operation with simple meal announcements instead of detailed carb counting.
- Considerations: Currently, there is limited scientific evidence supporting the system's safety and effectiveness. The TouchCare Nano CGM lacks peer-reviewed data to confirm its reliability. The lack of transparency in how the AI algorithm determines insulin delivery may conflict with requirements of the EU AI Act.

8. Twiist AID system
- Features: The Twiist system is the only AID system that can be controlled directly from an Apple Watch. The algorithm runs on the pump, so it remains in automode even when the phone is not nearby. It also allows for lower glucose target values than most other AID systems.
- Considerations: The Twiist AID system is currently available only for iPhone users in the United States. Although it is based on the DIY Loop algorithm, popular features such as overrides, Auto Bolus and remote commands are not included.

9. Open-Source AID Systems
- Features: AndroidAPS, DIY Loop, iAPS and Trio stand out with their full control over the insulin delivery, and the fact that you have a lot more choices of glucose sensor and insulin pumps.
- Considerations: It's important to consider that technical expertise is required, and that these systems are not regulated and should be used on your own risk. For

troubleshooting you can turn to the online community, as there is no official customer service.

The evolution in automated insulin delivery systems goes really fast. This content is updated in November 2025. For the latest updates, please visit the company's websites. Additionally, staying updated with resources like the Diabetotech blog can help you explore the evolving diabetes tech field.

It is essential to consult with your healthcare provider when selecting an automated insulin delivery system. Consider factors such as reimbursement availability, system compatibility with your diabetes centre, and the specific advantages and disadvantages of each system. Comparison websites and resources provided by your healthcare team can assist you in making an informed decision.

In the downloads section, you will also find a visually appealing PDF that highlights 10 Automated Insulin Delivery Systems and their key features.

By completing this module, you have demonstrated a commitment to understanding and optimizing your diabetes management. Acquiring knowledge about automated insulin delivery systems equips you with the tools to embrace the latest advancements in diabetes technology. You are now better prepared to engage in discussions with your healthcare provider, actively participate in treatment decisions, and achieve improved glycemic control.

Remember, diabetes management is a journey, and each step forward is a remarkable achievement. Continue to stay informed, seek support from your healthcare team, and embrace the opportunities that automated insulin delivery systems offer for a more manageable and fulfilling life with diabetes.

Congratulations once again on completing this module, and we wish you continued success in your diabetes management endeavours!